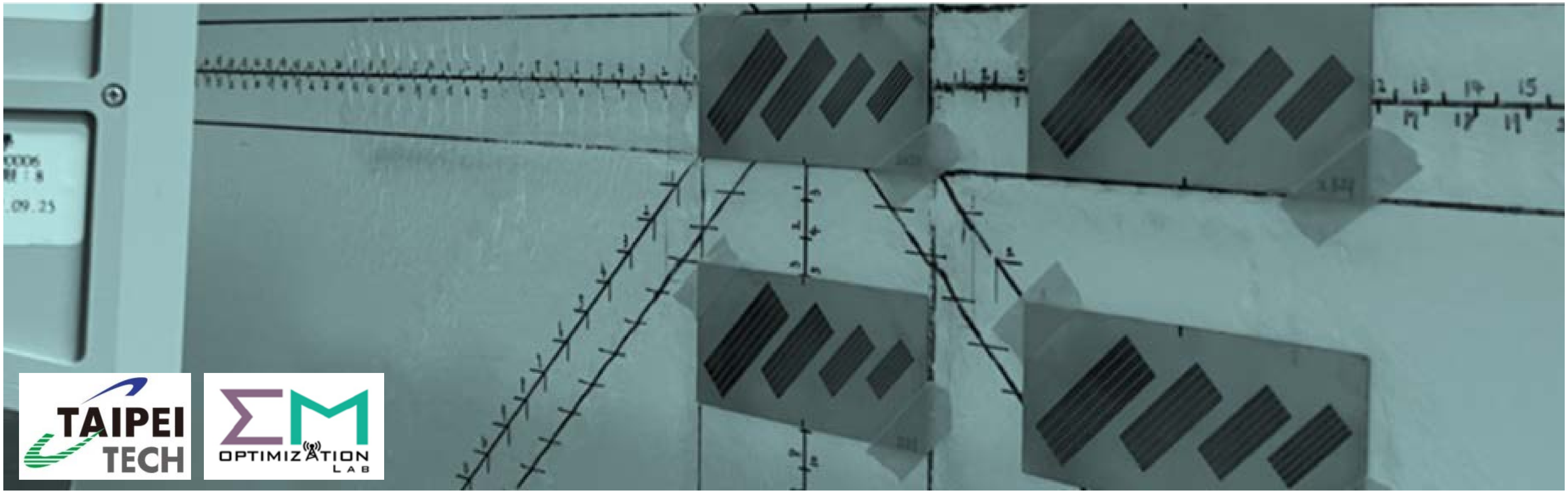
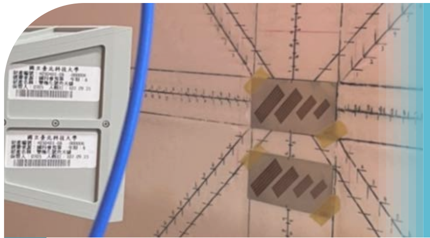
Linear Algebra
線性代數



Chapter 4: Determinants

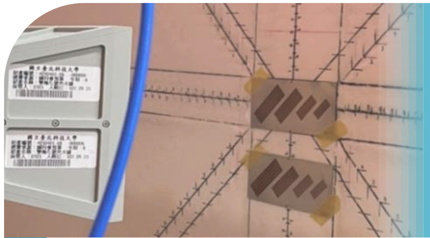
Yen-Sheng Chen (陳晏笙)

OCW Course Content



Agenda

1. **Properties of Determinants**
2. **Formulas for Determinants**
3. **Applications of Determinants**



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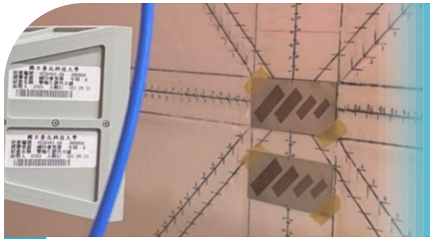


Introduction

$\mathbf{A}x = b$

Determinants

$\mathbf{A}x = \lambda x$

(only for squared $\mathbf{A}$)

- Existence of x
- Uniqueness of x
- Best x if unsolvable
- The most important tool: GE

- Diagonalization of $\mathbf{A}$
- Powers of $\mathbf{A}$
- Exponential of $\mathbf{A}$
- Special cases for $\mathbf{A}$
- A useful tool: Determinants

Introduction

An attempt to summarize a matrix!

To find pivots

To find $\mathbf{A}^{-1}$

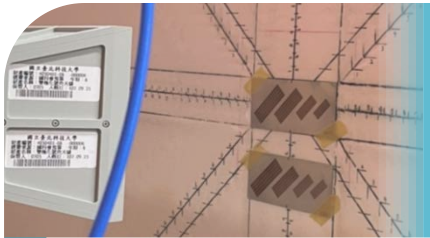
Test for invertibility

The determinant of

$$\mathbf{A} = \begin{bmatrix} * & * & \cdots & * \\ * & & & \vdots \\ \vdots & & & \vdots \\ * & * & \cdots & * \end{bmatrix}$$

To find x for $\mathbf{A}\mathrm{x} = \mathrm{b}$
(Cramer's rule)

The volume of a parallelepiped P



Three Basic Properties of Determinants

- Remember $\det\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$?

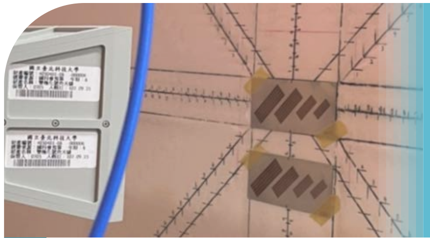
  This formula will be used to check the following properties

1. The determinant depends linearly on one row

$$\begin{vmatrix} a+a' & b+b' \\ c & d \end{vmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} + \begin{vmatrix} a' & b' \\ c & d \end{vmatrix} = \det \mathbf{B} + \det \mathbf{C} \neq \det\left(\mathbf{B}+\mathbf{C}\right)$$

That is, only one row is allowed to change

$$\begin{vmatrix} ta & tb \\ c & d \end{vmatrix} = t\begin{vmatrix} a & b \\ c & d \end{vmatrix} \neq \det\left(t\mathbf{A}\right) = t^2 \det\left(\mathbf{A}\right)$$



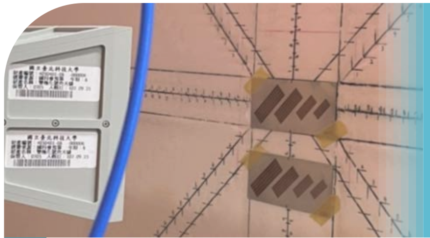
Three Basic Properties of Determinants

2. The determinant changes sign when two rows are exchanged

$$\begin{vmatrix} c & d \\ a & b \end{vmatrix} = cb - ad = -\begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

3. The determinant of $\mathbf{I}$ is 1

$$\begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1, \quad \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = 1, \quad \cdots$$



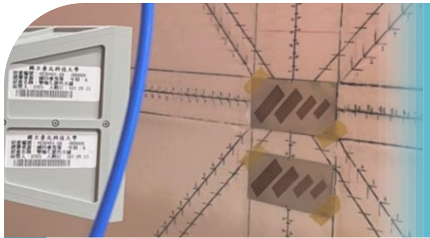
Derived Properties of Determinants

4. If two rows of $\mathbf{A}$ are equal, $\det\mathbf{A} = 0$

$$\begin{vmatrix} a & b \\ a & b \end{vmatrix} = 0$$

5. The row operations mentioned in Ch. 1 and Ch. 2 do not change the determinant

$$\begin{vmatrix} a+lc & b+ld \\ c & d \end{vmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$



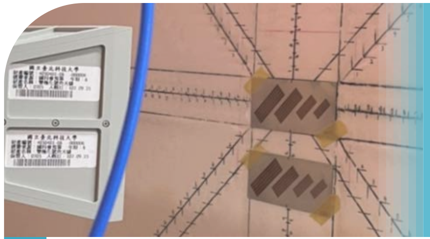
Derived Properties of Determinants

6. If $\mathbf{A}$ has a zero row, $\det\mathbf{A} = 0$

$$\begin{vmatrix} a & b \\ 0 & 0 \end{vmatrix} = 0$$

7. If $\mathbf{A}$ is triangular, $\det\mathbf{A}$ is the product of the entries on the main diagonal

$$\begin{vmatrix} a & b \\ 0 & d \end{vmatrix} = ad, \quad \begin{vmatrix} a & 0 \\ c & d \end{vmatrix} = ad.$$



Derived Properties of Determinants

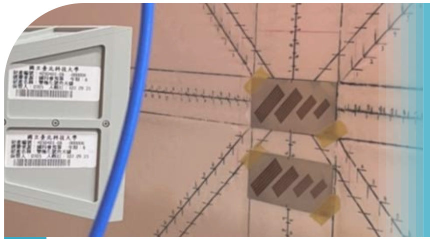
8. If $\mathbf{A}$ is singular, $\det\mathbf{A} = 0$.

   If $\mathbf{A}$ is invertible, $\det\mathbf{A} \neq 0$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \text{ is singular if and only if } ad - bc = 0$$

This property gives the first formula for determinants:

$$\det \mathbf{A} = \pm d_1 d_2 \cdots d_n \ \left(d_i \text{ denote pivots}\right)$$



Derived Properties of Determinants

9. For any two squared matrices $\mathbf{A}$ and $\mathbf{B}$, $\det(\mathbf{AB}) = \det\mathbf{A} \times \det\mathbf{B}$

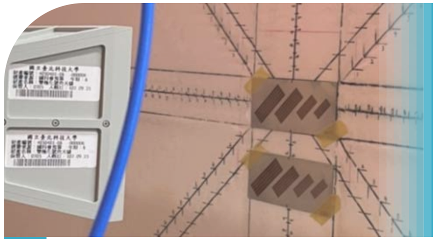
- Special case: $(\det \mathbf{A})\left(\det \mathbf{A}^{-1}\right) = \det\left(\mathbf{A}\mathbf{A}^{-1}\right) = \det \mathbf{I} = 1$

$$\Rightarrow \det \mathbf{A}^{-1} = \frac{1}{\det \mathbf{A}}$$

10. $\det\mathbf{A} = \det\mathbf{A}^{T}$

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = \begin{vmatrix} a & c \\ b & d \end{vmatrix}$$

- All the properties for rows are now applicable to columns!



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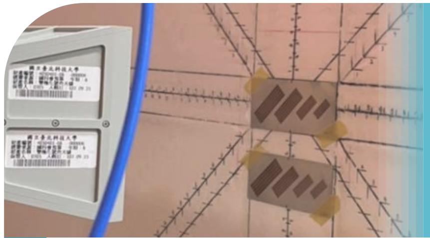


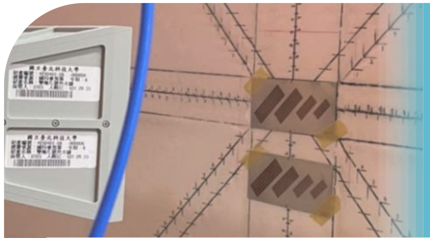
1st Formula for Determinants

- Determinants can be computed by the result of Gaussian elimination

$$\mathbf{A} = \mathbf{P}^{-1}\mathbf{LDU}$$

$$\det \mathbf{A} = \left(\det \mathbf{P}^{-1}\right)\left(\det \mathbf{L}\right)\left(\det \mathbf{D}\right)\left(\det \mathbf{U}\right)$$

$$\det \mathbf{A} = \pm\left(\det \mathbf{D}\right) = \pm d_1 d_2 \cdots d_n$$



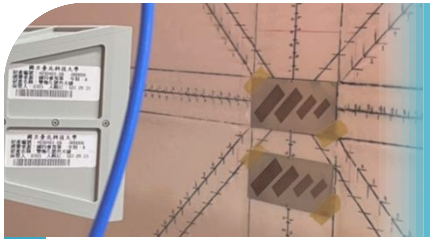
Examples

- Without row exchange:

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} =$$

- With row exchange:

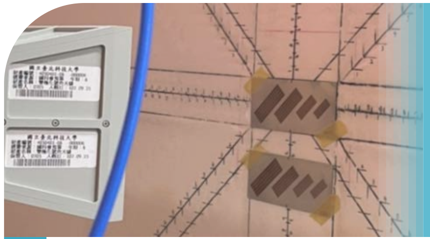
$$\det \mathbf{PA} = \det \begin{bmatrix} c & d \\ a & b \end{bmatrix} =$$



Examples

- Finite second-order difference matrix

$$\det\begin{bmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & -1 & 2 & -1 & \\ & & & \ddots & -1 \\ & & & -1 & 2 \end{bmatrix} =$$



2^{nd} Formula for Determinants

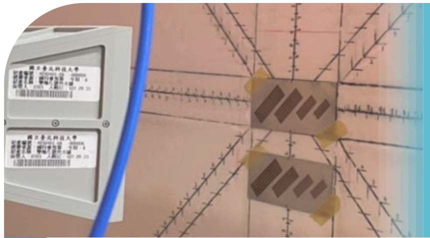
- The 1^{st} formula requires pivots
- Pivots come from $\mathbf{U}$, which comes from Gaussian elimination
- Can we get the determinant directly from the entries in $\mathbf{A}$?

- This requires breaking down every row in a matrix

$$\begin{bmatrix} a & b \end{bmatrix} = \begin{bmatrix} a & 0 \end{bmatrix} + \begin{bmatrix} 0 & b \end{bmatrix}$$

$$\Rightarrow \begin{vmatrix} a & b \\ c & d \end{vmatrix} = \begin{vmatrix} a & 0 \\ c & d \end{vmatrix} + \begin{vmatrix} 0 & b \\ c & d \end{vmatrix}$$

$$= \underbrace{\begin{vmatrix} a & 0 \\ c & 0 \end{vmatrix}}_{0} + \begin{vmatrix} a & 0 \\ 0 & d \end{vmatrix} + \begin{vmatrix} 0 & b \\ c & 0 \end{vmatrix} + \underbrace{\begin{vmatrix} 0 & b \\ 0 & d \end{vmatrix}}_{0}$$



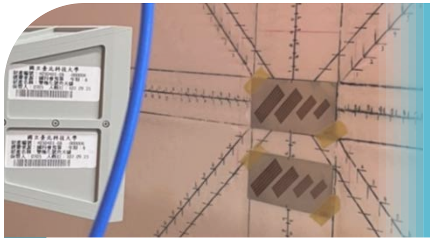
2^{nd} Formula for Determinants

- For any $n \times n$ matrix, each row can be split into n terms
- Combining n rows, how many terms are in the expansion?

- More importantly, how many terms survive?
- Only when the rows point in different directions, or nonzero terms come in different columns, this individual term survive!

- Example: 3×3 case

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} =$$



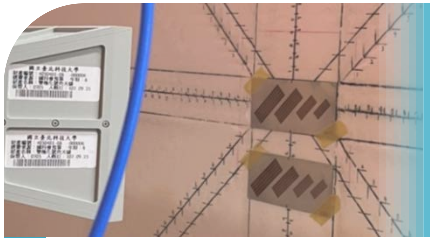
2nd Formula for Determinants

- When the 1st row has a nonzero entry in column α, the 2nd row must have a nonzero entry in different column β, and so on till finally the nth row is nonzero in column v
- All these nonzero column numbers should be different ($\alpha \neq \beta \neq \ldots \neq v$)
- This is like the permutation of numbers
- Example: 3×3 case

$$(\alpha, \beta, v) = (1, 2, 3), (2, 3, 1), (3, 1, 2), (1, 3, 2), (2, 1, 3), (3, 2, 1)$$

$$\det \mathbf{A} = a_{11}a_{22}a_{33}\begin{vmatrix} 1 & & \\ & 1 & \\ & & 1 \end{vmatrix} + a_{12}a_{23}a_{31}\begin{vmatrix} & 1 & \\ & & 1 \\ 1 & & \end{vmatrix} + a_{13}a_{21}a_{32}\begin{vmatrix} & & 1 \\ 1 & & \\ & 1 & \end{vmatrix}$$

$$+ a_{11}a_{23}a_{32}\begin{vmatrix} 1 & & \\ & & 1 \\ & 1 & \end{vmatrix} + a_{12}a_{21}a_{33}\begin{vmatrix} & 1 & \\ 1 & & \\ & & 1 \end{vmatrix} + a_{13}a_{22}a_{31}\begin{vmatrix} & & 1 \\ & 1 & \\ 1 & & \end{vmatrix}$$



2nd Formula for Determinants

$$\det \mathbf{A} = \sum_{\sigma} \underbrace{\left(a_{1\alpha} a_{2\beta} \cdots a_{n\nu}\right) \det \mathbf{P}_{\sigma}}_{n!\ \text{terms}}$$

- $\det\mathbf{P}_\sigma = +1$ or -1, depending on whether the number of exchanges is even or odd
- $(\alpha, \beta, \nu) = (1, 3, 2) \rightarrow \det\mathbf{P}_\sigma = -1$
- $(\alpha, \beta, \nu) = (3, 1, 2) \rightarrow \det\mathbf{P}_\sigma = (-1)^2 = 1$

- Check: 2×2 case

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} =$$

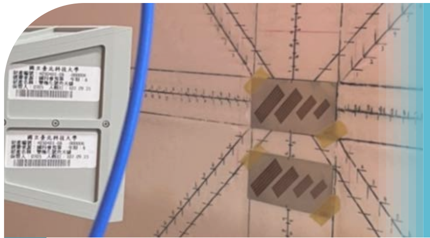
Cofactors

- To make the formula more compact, a new notation is defined:

$$\det \mathbf{A} = \sum_{\sigma}\left(a_{1\alpha}a_{2\beta}\cdots a_{n\nu}\right)\det \mathbf{P}_{\sigma} = \sum_{\alpha=1}^{n} a_{1\alpha}\underbrace{\sum_{\sigma'}\underbrace{\left(a_{2\beta}\cdots a_{n\nu}\right)\det \mathbf{P}_{\sigma'}}_{(n-1)!\text{ terms}}}_{C_{1\alpha}}$$

- σ = (α, β, …, ν) are n! permutations
- $\mathbf{P}_{\sigma'}$ is a (n–1) × (n–1) permutation matrix
- $C_{1\alpha}$: the cofactor of the 1st row and α^{th} column

$$\Rightarrow \quad \det \mathbf{A} = \sum_{\alpha=1}^{n} a_{1\alpha}C_{1\alpha} = a_{11}C_{11} + a_{12}C_{12} + \cdots + a_{1n}C_{1n}$$

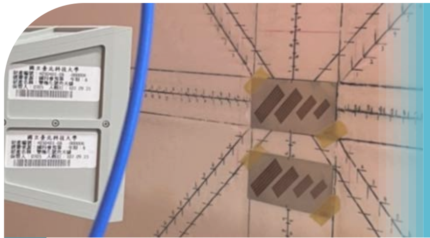


Calculation of Cofactors

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} a_{11} & & \\ & a_{22} & a_{23} \\ & a_{32} & a_{33} \end{vmatrix} + \begin{vmatrix} & a_{12} & \\ a_{21} & & a_{23} \\ a_{31} & & a_{33} \end{vmatrix} + \begin{vmatrix} & & a_{13} \\ a_{21} & a_{22} & \\ a_{31} & a_{32} & \end{vmatrix}$$

$$= a_{11}\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12}\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13}\begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

- If matrix $\mathbf{M}_{1j}$ is the matrix formed by throwing away row 1 and column j, then $C_{1j} = (-1)^{1+j} \det\mathbf{M}_{1j}$



2nd Formula for Determinants

- The expansion can be conducted for any row i:

$$\det \mathbf{A} = \sum_{\alpha=1}^{n} a_{i\alpha} C_{i\alpha} = a_{i1}C_{i1} + a_{i2}C_{i2} + \cdots + a_{in}C_{in}$$

where cofactor C_{ij} is the determinant of $\mathbf{M}_{ij}$ with correct sign:

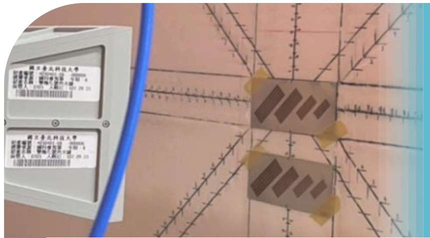
$$C_{ij} = (-1)^{i+j} \det \mathbf{M}_{ij}$$

where $\mathbf{M}_{ij}$ is formed by deleting row i and column j of $\mathbf{A}$

- Example: 4 × 4 finite difference matrix

$$\det \begin{bmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & -1 & 2 & -1 \\ & & -1 & 2 \end{bmatrix} =$$

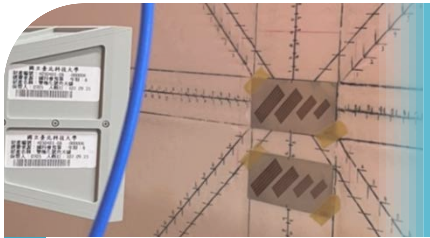
Question: Which row should be expanded?



Examples

- Consider $\mathbf{A} = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$,
  - ◆ find the determinant of $\mathbf{A}$
  - ◆ Find the determinant of $\mathbf{A} + \mathbf{I}$
  - ◆ Find the determinant of $\mathbf{A} + 2\mathbf{I}$

- This problem is about the n by n matrix $\mathbf{A}_n$ that has zeros on its main diagonal and all other entries equal to -1. Find the determinant of $\mathbf{A}_n$
- Hint: Start by adding all rows (except the last) to the last row, and then factoring out a constant. You could check $n = 3$ to have a start



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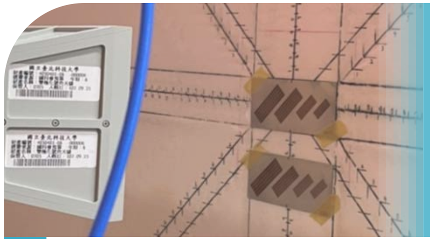


Applications

A tool for computing eigenvalues

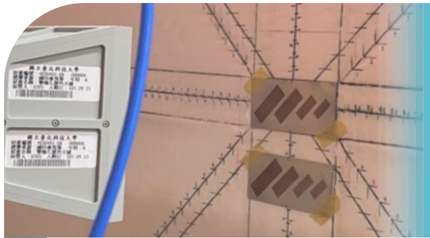
in chapter 5

Computation of A^{-1}

Cramer's rule for solving $Ax = b$

Volume of a box

Formula for pivots



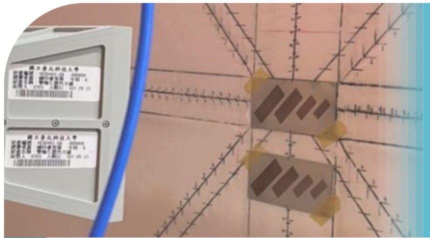
Computation of $\mathbf{A}^{-1}$

- For 2×2 case, the $\mathbf{A}^{-1}$ is related to cofactors:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \frac{1}{\det \mathbf{A}}\begin{bmatrix} C_{11} & C_{21} \\ C_{12} & C_{22} \end{bmatrix}$$

$$\mathbf{A}^{-1} = \frac{\mathbf{C}^T}{\det \mathbf{A}} \text{ or } \left(\mathbf{A}^{-1}\right)_{ij} = \frac{C_{ji}}{\det \mathbf{A}}$$

- **C**: Cofactor matrix → $\begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1n} \\ C_{21} & & & C_{2n} \\ \vdots & & & \vdots \\ C_{n1} & C_{n2} & \cdots & C_{nn} \end{bmatrix}$



Examination

- This is to show $\mathbf{AC}^T = (\det\mathbf{A})\mathbf{I}$, or

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & & & a_{2n} \\ \vdots & & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}\begin{bmatrix} C_{11} & C_{21} & \cdots & C_{n1} \\ C_{12} & & & C_{n2} \\ \vdots & & & \vdots \\ C_{1n} & C_{2n} & \cdots & C_{nn} \end{bmatrix} = \begin{bmatrix} \det\mathbf{A} & 0 & \cdots & 0 \\ 0 & \det\mathbf{A} & & 0 \\ \vdots & & \det\mathbf{A} & \vdots \\ 0 & 0 & \cdots & \det\mathbf{A} \end{bmatrix}$$

- Why do we get $\det\mathbf{A}$ on the diagonal?
- Why do we get zeros off the diagonal?
- For row 1 of $\mathbf{A}$ and column 2 of $\mathbf{C}$

$$a_{11}C_{21} + a_{12}C_{22} + \cdots + a_{1n}C_{2n} = \det\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} =$$

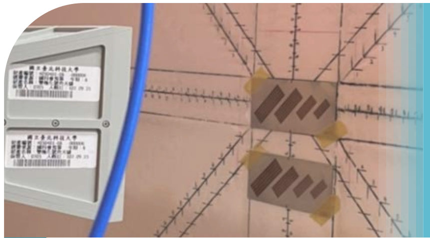
Examples

- 2×2 case

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} =$$

- The inverse of

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$



Test of Singularity

Find $\mathbf{A}^{-1}$

From Gauss-Jordan:

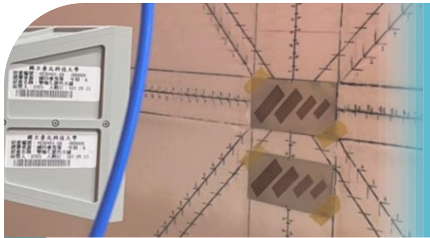
$$\mathbf{L}^{-1}\begin{bmatrix}\mathbf{A} & \mathbf{I}\end{bmatrix} = \begin{bmatrix}\mathbf{U} & \mathbf{L}^{-1}\end{bmatrix}$$

$$\mathbf{U}^{-1}\begin{bmatrix}\mathbf{U} & \mathbf{L}^{-1}\end{bmatrix} = \begin{bmatrix}\mathbf{I} & \mathbf{A}^{-1}\end{bmatrix}$$

From determinant:

$$\mathbf{A}^{-1} = \frac{1}{\det \mathbf{A}}\mathbf{C}^{T}$$

- Singularity of a squared matrix $\mathbf{A}$:
  - Linearly dependent columns
  - Linearly dependent rows
  - The number of pivots $r < n$
  - Dimension of C($\mathbf{A}$) $< R^n$
  - Dimension of C($\mathbf{A}^T$) $< R^n$
  - Non-invertible $\mathbf{A}$
  - $\mathbf{A}$ has a nontrivial nullspace
  - $\mathbf{A}$ has a nontrivial left nullspace
  - $\det\mathbf{A} = 0$
  - $n - r$ repeated zero eigenvalues (in Ch. 5)



Cramer's Rule for Solving $\mathbf{Ax} = \mathbf{b}$

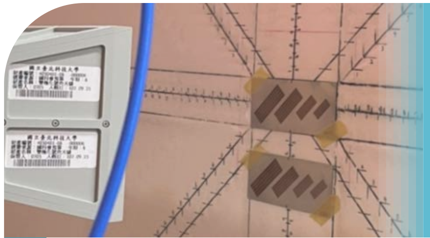
$$\mathbf{A}\mathrm{x} = \mathrm{b}$$

$$\mathrm{x} = \mathbf{A}^{-1}\mathrm{b} = \frac{1}{\det \mathbf{A}}\mathbf{C}^T\mathrm{b} \Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_j \\ \vdots \\ x_n \end{bmatrix} = \frac{1}{\det \mathbf{A}} \begin{bmatrix} C_{11} & C_{21} & \cdots & C_{n1} \\ C_{12} & C_{22} & & C_{n2} \\ C_{1j} & C_{2j} & & C_{nj} \\ \vdots & \vdots & & \vdots \\ C_{1n} & C_{2n} & \cdots & C_{nn} \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ b_j \\ \vdots \\ b_n \end{bmatrix}$$

Be aware of the subscripts!

- $\mathbf{C}^T\mathrm{b}$ is a vector. x_j is expressed as: $x_j = \frac{1}{\det \mathbf{A}}\left(C_{1j}b_1 + C_{2j}b_2 + \cdots + C_{nj}b_n\right)$
- Based on the reverse usage of the determinant, this is like:

$$x_j = \frac{1}{\det \mathbf{A}} \det \begin{bmatrix} a_{11} & a_{21} & \cdots & a_{n1} \\ a_{12} & a_{22} & & a_{n2} \\ b_1 & b_2 & & b_n \\ \vdots & \vdots & & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{nn} \end{bmatrix} = \frac{1}{\det \mathbf{A}} \det \begin{bmatrix} a_{11} & a_{12} & b_1 & \cdots & a_{1n} \\ a_{21} & a_{22} & b_2 & & a_{2n} \\ \vdots & \vdots & \vdots & & \\ a_{n1} & a_{n2} & b_n & \cdots & a_{nn} \end{bmatrix}$$



Volume of a Box

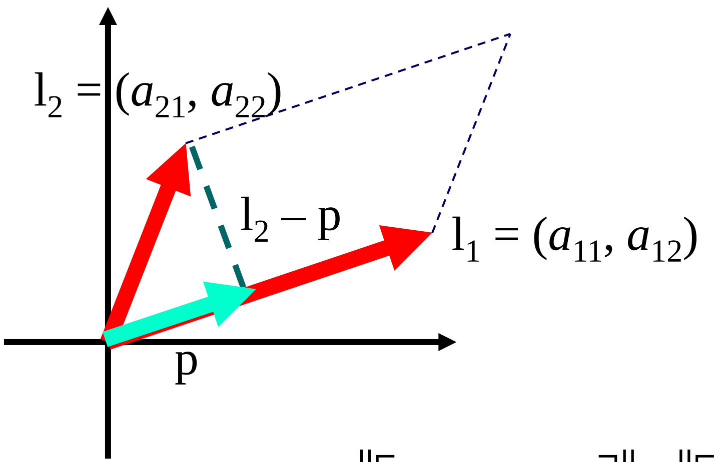
- Considering a parallelepiped, the lengths of edges are denoted by $l_1, l_2, \ldots, l_n$ (vectors)
- If all angles are right angles, the volume is $l_1 \times l_2 \times l_3 \times \ldots \times l_n$
- Now, let's put the lengths as the row of $\mathbf{A}$:

$$\mathbf{A}\mathbf{A}^T = \begin{bmatrix} \text{row } 1\ (l_1) \\ \text{row } 2\ (l_2) \\ \vdots \\ \text{row } n\ (l_n) \end{bmatrix} \begin{bmatrix} \mathrm{r} & \mathrm{r} & & \mathrm{r} \\ \mathrm{o} & \mathrm{o} & & \mathrm{o} \\ \mathrm{w} & \mathrm{w} & \cdots & \mathrm{w} \\ 1 & 2 & & n \\ (l_1) & (l_2) & & (l_n) \end{bmatrix} = \begin{bmatrix} l_1^2 & 0 & \cdots & 0 \\ 0 & l_2^2 & & 0 \\ \vdots & & & \vdots \\ 0 & 0 & \cdots & l_n^2 \end{bmatrix}$$

$$\Rightarrow \det\left(\mathbf{A}\mathbf{A}^T\right) = \det(\mathbf{A})\det\left(\mathbf{A}^T\right) = (\det \mathbf{A})^2 =$$

$$\Rightarrow \left|\det \mathbf{A}\right| =$$

Non-Right Angle Case



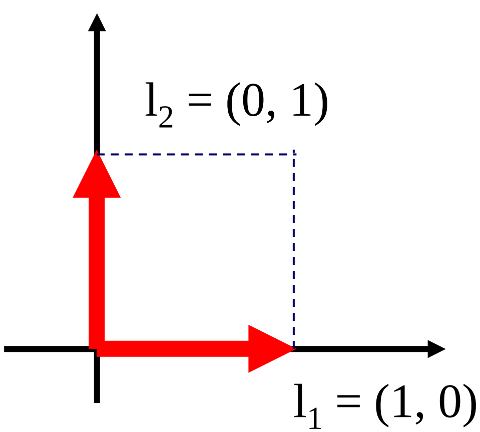
$$\text{Area} = \|l_2 - p\| \times \|l_1\| = \|l_2 - x l_1\| \times \|l_1\| = \underbrace{\left\| \begin{bmatrix} a_{21} - x a_{11} \\ a_{22} - x a_{12} \end{bmatrix} \right\| \times \left\| \begin{bmatrix} a_{11} \\ a_{12} \end{bmatrix} \right\|}_{\text{Back to the right angle case!}}$$

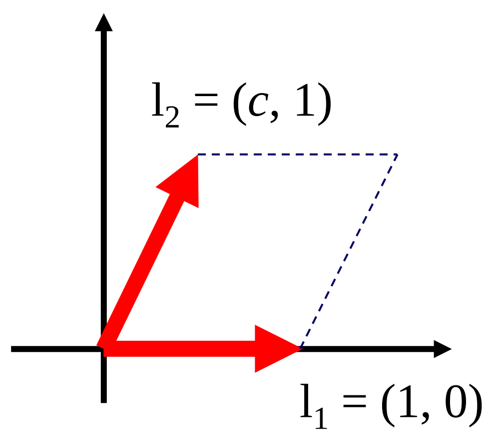
$$\Rightarrow \text{Area} = \left| \det \begin{bmatrix} a_{21} - x a_{11} & a_{22} - x a_{12} \\ a_{11} & a_{12} \end{bmatrix} \right| = \left| \det \begin{bmatrix} a_{21} & a_{22} \\ a_{11} & a_{12} \end{bmatrix} \right| = \left| \det \mathbf{A} \right|$$

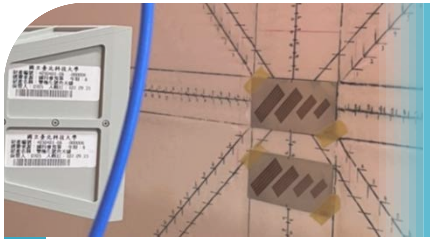
Volume of a Box (Summary)

- Row operations do not change the determinant
- For both right angle and non-right angle cases, the volume is determined by $|\det \mathbf{A}|$
- Gram-Schmidt process is a form of row operations
- Gram-Schmidt process without dividing the length does not change the determinant

- Example





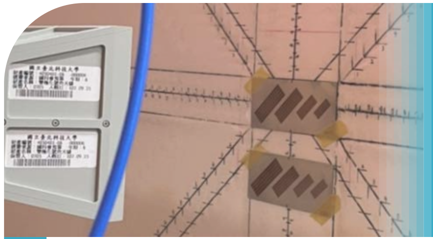


Formula for Pivots

$$\mathbf{A} = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \rightarrow \mathbf{A} = \mathbf{LDU}$$

- What does the first pivot depend on?
- What does the second pivot depend on?

- If $\mathbf{A}$ is factored into $\mathbf{LDU}$, the upper left corners satisfy $\mathbf{A}_k = \mathbf{L}_k\mathbf{D}_k\mathbf{U}_k$
- For every k, the submatrix $\mathbf{A}_k$ is going through a Gaussian elimination of its own



Formula for Pivots

$$\mathbf{A}_k = \mathbf{L}_k \mathbf{D}_k \mathbf{U}_k$$

$$\rightarrow \det \mathbf{A}_k = \left(\det \mathbf{L}_k\right)\left(\det \mathbf{D}_k\right)\left(\det \mathbf{U}_k\right) = \det \mathbf{D}_k = d_1 d_2 \cdots d_n$$

Numerical method for finding pivots

$$d_1 = \det\left(\mathbf{A}_1\right) = a_{11}$$

$$d_2 = \frac{\det\left(\mathbf{A}_2\right)}{\det\left(\mathbf{A}_1\right)} \qquad \cdots \qquad d_k = \frac{\det\left(\mathbf{A}_k\right)}{\det\left(\mathbf{A}_{k-1}\right)}$$

- The pivot entries are all nonzero whenever the numbers $\det\mathbf{A}_k$ are all nonzero
- Elimination can be completed without row exchanges if and only if the leading submatrices $\mathbf{A}_1, \mathbf{A}_2, \ldots, \mathbf{A}_n$ are all nonsingular